

Reproducing the pipeline in Python

A version-independent script that regenerates every calculated tab —
and matches the workbook to machine precision

The goal

The Excel workbook reduces raw mass-spec beams into noble-gas concentrations and isotope ratios across several tabs. We rebuild every calculated tab in Python — for **any monthly version** of the workbook — and prove it reproduces the original.

01

Reproduce

Regenerate Semi Raw, Blanks, Air STDs and Samples as worksheets.

02

Generalize

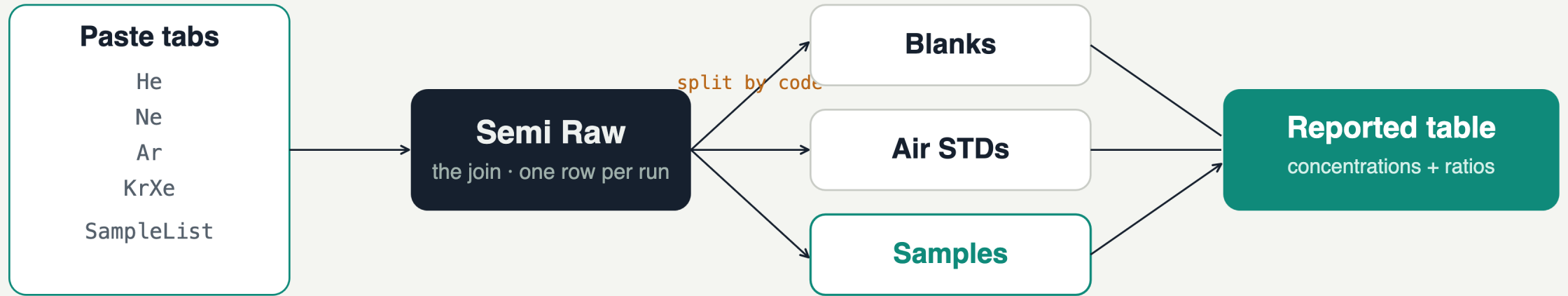
Read the layout & constants from the file, so it survives sheet revisions.

03

Validate

Diff every regenerated cell against the workbook's own values.

The pipeline



Raw beams and the run log join into **Semi Raw**; the sample-type code routes each row to **Blanks**, **Air STDs** or **Samples**, where the reduction produces the reported values.

One idea, applied everywhere

Measure each sample **relative to the bracketing air standard**, then scale by a **known atmospheric value**. Everything else layers on top.

```
conc = (sample - blank) / (airstd - blank)
      × C_air × depletion × (split / line) / manometer
```

Ratio to air

blank-subtracted, vs
the standard

C_{air}

known ppm-in-air of
the gas

Depletion

air-reservoir draw-
down

Split / line

how much gas was
measured

Manometer

amount of gas
admitted

It reads the layout from the file

Nothing is hard-coded. For each output column, the script **parses its own formula** to discover the wiring, then reads the constants straight from their cells.

Beam column

which isotope feeds the cell

Blank & air-std cells

the reference rows it divides by

Abundance constant

ppm-in-air, wherever it lives

Split & line tables

read from their lookup ranges

Depletion on / off

detected from the formula text

Blank / air selection

expands any AVERAGE() range

What changed between versions

The same method — but Aug 2024 and Oct 2026 differ in almost every mechanical detail. The script adapts to all of it.

Aspect	Aug 2024	Oct 2026
Output columns	BO:BY	BJ:CN
Abundance constants	fixed cells	live refs, recalibrated
Split / line tables	one calibration	different values, moved
Depletion term	applied	dropped
Blank selection	3-row range	single row
Air-std bracket	contiguous	non-contiguous

Aug 2024 → Oct 2026

Architecture: one class per tab

GasWorkbook

loads the file · builds Semi Raw · computes shared blank / air-std references · copies every tab through

SemiRawTab

validates the join; keeps
manual overrides

BlanksTab

reproduces the avg row

AirStdTab

blank-corrected + ratio
columns

SamplesTab

concentrations + ratios

Copy everything through, overlay the recomputed columns. Reference data and tabs are preserved as-is; only the calculated columns are recomputed — and diffed.

GENERATED OUTPUT vs SOURCE WORKBOOK

1,177,149 cells.
0 real differences.

The Python reproduction matches the original workbook to machine precision — across all ten tabs.

Validation

743,207

cells exactly equal

57,133

differ only in the 15th digit

0

cells dropped or added

0

text / label mismatches

1.4×10^{-15}

worst relative difference anywhere — the limit of double-precision arithmetic, not a real discrepancy. Every one of the 10 tabs agrees.

Edge cases it handles

Manual overrides

a hand-set cell (Semi Raw!!880) is read from Semi Raw, not re-joined from pastes

Broken columns

#REF! columns in the sheet are detected and skipped

Missing abundance

an isotope with no constant resolves to 0, matching the sheet

LOOKUP semantics

string-typed codes + Excel's approximate-match clamping are replicated

Scattered selections

non-contiguous AVERAGE() arguments are expanded exactly

Sheet anomalies

the workbook's own hand-edited rows are flagged, not silently matched

What you get

`output_tabs.py`

one OOP script that runs on any version of the workbook

`--separate`

one combined workbook, or one .xlsx file per tab

`validated`

every calculated column recomputed and diffed, tab by tab

Exact reproduction, to machine precision — portable across the sheet's monthly revisions.